



A new model for pressure drop correction for series-arranged misaligned monoliths

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ARTICLE INFO

Keywords:
Monoliths
Pressure drop
Misalignment
CFD
Channels

ABSTRACT

This study presents a new model to predict the additional pressure drop for series-arranged misaligned monoliths, relevant for high-flow industrial applications. Long reactor tubes often contain multiple short monoliths, leading to extra pressure drops as the flow enters and leaves multiple monoliths. Existing models address aligned monoliths, but this research considers misalignment, where extra pressure drop is higher. Channel Reynolds from 50 to 320, spacings from 0.1 to 20 channel diameters, and several void fractions are investigated using a discrete channel model. Results indicate a significant impact of misalignment on extra pressure drops in closely spaced monoliths due to flow collisions. A correction factor model that adapts to misalignment is proposed, starting from large values for closely spaced channels and decreasing as spacing increases. The proposed model aligns with discrete channel data and phenomenological behavior of aligned channels both in trend and magnitude, showing an R^2 above 0.93.

1. Introduction

Monoliths consist of solid pieces with parallel straight channels that are widely used as catalytic support (Williams, 2001; Lock et al., 2021; Hayes and Cornejo, 2021). That shape gives high surface area while maintaining a compact structure, also good structural integrity, low backpressure and good flow distribution (Önsan and Avci, 2016; Cornejo and Hayes, 2021; Ibáñez et al., 2021). Initially, monoliths were used as single substrates in containers (Della Torre et al., 2020; Tischer et al., 2001; Tronconi and Forzatti, 1992). Later, it became more common to find applications implying more than one monolith in series (Chatterjee et al., 2009; Arab et al., 2014; Jang et al., 2018). Monoliths have gained attention during the last decade because they are particularly suitable for high-flow applications, such as automotive exhaust gas cleaning, CO₂ hydrogenation to methanol, reforming, and propylene ammoxidation to acrylonitrile, to name a few (Arab et al., 2014; Azzoni et al., 2022; Salamon et al., 2020; Tahay et al., 2018; Schädel et al., 2009). Additive manufacturing has enabled the possibility of proposing novel monolith shapes and compositions. For example, Quintanilla et al. (2018) reported a 3D-printed Fe-doped silicon carbide monolithic catalyst for wet peroxide oxidation processes, Busse et al. (2018) compared open-cell structures made with additive manufacture with monoliths, and Negri et al. (2018) tested several novel monolith shapes for monopropellant thrusters. Our group has also been testing and modeling flow

through monoliths with a non-homogeneous channel size (Cornejo et al., 2021; Reinao and Cornejo, 2023), while Kane and Northrop (2023) studied heat transfer in such configurations.

In monoliths, the inner walls of the channels are usually coated with a catalyst layer a few microns thin called washcoat. Washcoat thickness prevents internal diffusion limitations, while external mass transfer limitations are usually avoided by selecting small channel hydraulic diameters, typically ranging from 0.5 to 2 mm (Ekström and Andersson, 2002; Clarkson, 1997). The most typical channel shape is square, however, it is possible to find monoliths with circular, triangular or hexagonal channels (Groppi and Tronconi, 1997; Depcik and Hausmann, 2013; Ding et al., 2023). Commonly, monoliths are made of ceramic materials, but metallic and silicon carbide substrates have also been tested (Quintero et al., 2022; Mehdipour et al., 2024). Another option is corrugated metallic monoliths whose channel shape is usually assumed to be triangular (Avila et al., 2005; Tronconi et al., 2004; Shen et al., 2024). It is worth mentioning that for all channel shapes, it has been observed that, after washcoating, the channel tends to become rounder (Watling, 2020; Peszynski and Tesař, 2020).

Inlet and outlet effects in monoliths, usually taken as minor losses, become particularly important in high-flow applications. The pressure drop from these effects scales quadratically with flow rate, unlike other losses, which scale linearly (Quadri et al., 2010; Carter et al., 1998). This has also been observed for the so-called wall-flow monoliths used in

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<https://doi.org/10.1016/j.ces.2024.120515>

Received 28 May 2024; Received in revised form 5 July 2024; Accepted 15 July 2024

Available online 24 July 2024

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particulate filters, where, depending on the operating condition, may become relevant (Mesquida et al., 2020; Watling et al., 2021; Mitsouridis et al., 2023). That underscores the relevance of inlet and outlet effects in the applications where monoliths offer the most advantages.

In a physical system, experimentally measuring inlet and outlet losses is quite challenging. On the other hand, discrete channel computational fluid dynamics (CFD) models, such as those used in this paper, stand out for their ability to accurately simulate fluid dynamics within the narrow confines of monolith channels. These models are particularly useful for studying channel-scale phenomena, where the physical insertion of probes can disrupt the very flow fields that are wanted to measure (Hettel et al., 2013). In the past, CFD has been successfully utilized to describe channel scale phenomena in monoliths, with many works reported in the literature for cold and reacting flow (Iwaniszyn et al., 2021; Canu and Vecchi, 2002; Cornejo et al., 2022; Pawlowski et al., 2018).

In general terms, pressure drop through a monolith can be computed as in Equation (1). In the equation, the terms Δp_{el} and Δp_{fd} occur inside of the channels and come from friction at the hydraulic entrance length and fully developed region. For simplicity, let us define the losses directly inside the channel as $\Delta p_c = \Delta p_{el} + \Delta p_{fd}$. Δp_i represents energy dissipation because of the upstream flow colliding with the frontal face of the substrate and entering the channels, while Δp_o are losses because of the flow leaving the substrate to an open section (Cornejo et al., 2020a). Among the aforementioned, Δp_{fd} is the only one that scales linearly with flow rate.

$$\Delta p = \underbrace{\Delta p_i}_{\text{entering}} + \underbrace{\Delta p_{el} + \Delta p_{fd}}_{\text{inside } (= \Delta p_c)} + \underbrace{\Delta p_o}_{\text{leaving}} \quad (1)$$

Early models assumed fully developed laminar flow and neglected any inlet, entrance and outlet effect; hence, the pressure drop through the monoliths was computed by using the Hagen-Poiseuille equation (Benjamin et al., 1996). Mixing the latter with Darcy's law made it possible to perform reactor scale simulations through the continuum approach since discrete channel models of a whole reactor are prohibitive for systematic research (Hettel et al., 2019; Bertrand et al., 2012). Quadri et al. (2010) reported a series of experiments with long and short monoliths, showing that depending on length, void fraction and flow rate, the so-called minor losses dominate at a point that the relationship between pressure drop and flow rate is no longer linear, it turns quadratic. Among the variables potentially explaining that non-linearity, upstream turbulence entering the channels was reported to have no effect on pressure drop (Strom et al., 2011; Cornejo et al., 2020a; Garretón et al., 2023). Extra losses due to the entrance length were investigated by Shah and London (1978), and the effect of the flow entering at an angle into the channels was studied and reported by Benjamin et al. (2003). More recently, Cornejo et al. (2019) used a discrete channel model to compare losses at the different parts of a channel, concluding that, depending on flow rate and void fraction, the most relevant quadratic losses come from Δp_o , Δp_i and Δp_{el} , with Δp_o being the highest. Together can represent more than 30% of the total pressure drop. While such a percentage depends on channel length, this paper considers the same range of Reynolds numbers and a monolith with similar void fraction and channel size as in the cited study; hence, the same trend should also be valid.

Some applications require two or more monoliths in series. Examples are automotive gas after treatment, with a catalytic oxidation monolith followed by a selective reduction one, and methanol synthesis carried out in tube-shell reactors (Arab et al., 2014). The latter is highly exothermic; hence, the tubes are filled with catalyst and externally cooled. Those reactors can have thousands of tubes, several meters long each (Montebelli et al., 2014). Given the manufacturing limitations of long monolithic structures, such reactor tubes can contain several monoliths arranged in series, as shown in Fig. 1(a).

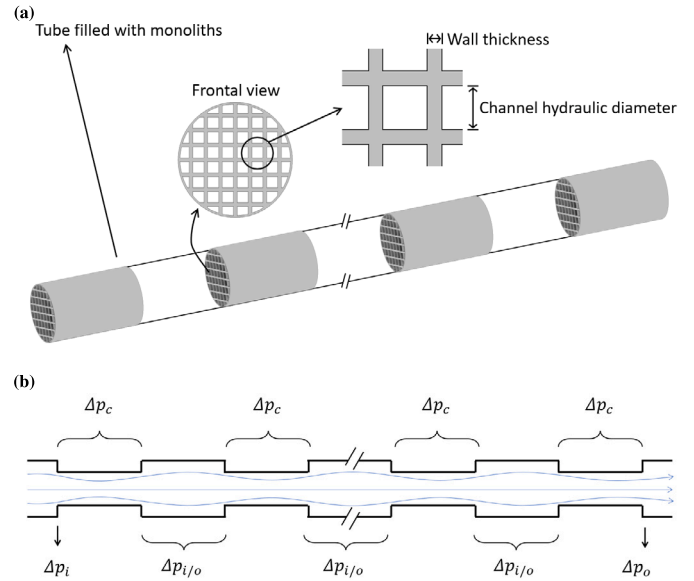


Fig. 1. (a) Reactor tube containing several monoliths. (b) Monolith channels in series.

A closer look at the monolith channels in an in-series system is shown in Fig. 1(b). It can be seen that, depending on the relative gap length between substrates, Equation (1) may not apply since the flow abandoning the first monoliths may encounter the second substrate instead of freely flowing in a large open area. On the other hand, the flow approaching the second monolith may not have a flat velocity profile. In such a case, Δp_o from one monolith combines with Δp_i of the monolith immediately downstream, being both terms represented as a single one named $\Delta p_{i/o}$. With that, Equation (1) can be adapted to describe the pressure drop through the whole system containing N monoliths, as follows:

$$\Delta p = \Delta p_i + N \Delta p_c + (N - 1) \Delta p_{i/o} + \Delta p_o \quad (2)$$

Expressions for Δp_i and Δp_o are available in the literature for several shapes (Cornejo et al., 2020b). The pressure drops for single monoliths can be computed using Equation (3) and (4).

$$\Delta p_i = f_1 \Big|_i \cdot \mu \cdot u_c + f_2 \Big|_i \cdot \frac{1}{2} \rho \cdot u_c^2 \quad (3)$$

$$\Delta p_o = f_2 \Big|_o \cdot \frac{1}{2} \rho \cdot u_c^2 \quad (4)$$

In Equation (3) and (4), f_1 and f_2 are empirical coefficients that depend on the monolith geometry and flow condition. f_1 corresponds to the viscous losses related to the fluid's viscosity, while f_2 to the inertial losses, which are proportional to the dynamic pressure. The subscripts i and o indicate the inlet and outlet faces of the substrate, respectively.

According to previous works (Cornejo, 2021), $\Delta p_{i/o}$ can be computed from the previous two as in Equation (5).

$$\Delta p_{i/o} = f_\delta (\Delta p_i + \Delta p_o) \quad (5)$$

The term f_δ is a correction factor that depends on the dimensionless spacing between substrates (δ^*). For perfectly aligned channels, it can be estimated using Equation (6). In the equation, τ_c is a model constant, D is channel diameter, Re is channel Reynolds number, and the group $(\delta/D \cdot Re)$ is δ^* .

$$f_\delta = 1 - \exp \left[-\tau_c \left(\frac{\delta}{D \cdot Re} \right) \right] \quad (6)$$

Equation (6) captures the dynamics for scenarios involving perfectly aligned monolith substrates. In such configurations, the longitudinal axes of the channels in a monolith align with those of the subsequent

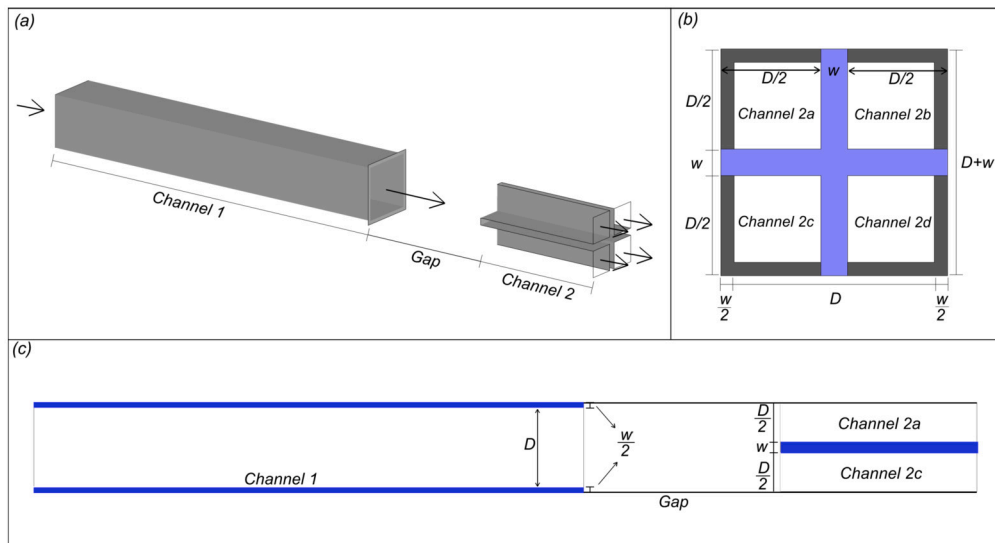


Fig. 2. Computational domain. (a) Both channels in series, (b) frontal view and (c) side view.

one. As a result, when these substrates are placed very closely, the viscous core flowing from the first channel directly enters into the next, making $\Delta p_{i/o}$ disappear because both channels behave as a single one but larger. In line with that, Equation (6) starts from 0 at a very close distance between substrates and grows to the asymptotic value of 1 when the substrates are so far from each other that Δp_o and Δp_i can be considered independent phenomena for each monolith.

In practical scenarios, achieving perfect alignment of monolith channels along the entire length of a tube may be uncommon. When misalignment occurs between consecutive monoliths, fluid flow and pressure drop dynamics exhibit notable differences. As the gap between misaligned monoliths narrows, an increase in pressure drop is observed. This phenomenon could be explained by the direct collision of the viscous core exiting one channel with the frontal wall or the inter-channel space of the subsequent monolith. In contrast with the aligned case, where the correction factor gradually increases from zero, in a misaligned situation, the correction factor should start from a higher initial value, then progressively diminish towards the asymptotic value of 1 as the spacing between the monoliths increases.

Currently, to the best of our knowledge, there are no comprehensive data or models available able to accurately capture the dynamics and quantify the correction factor, f_δ , in scenarios involving misaligned monolith channels. Focused on that knowledge gap, this paper uses a discrete channel computational model to investigate the flow field and pressure drop for misaligned monolith channels. A parametric study encompassing a range of channel Reynolds numbers and monolith spacing lengths is performed. This approach was specifically designed to span a broad spectrum of δ^* values, allowing for observing both the initial and asymptotic values of the correction factor for pressure drop between substrates in the context of misalignment. The findings are presented in terms of the flow field and pressure drop, along with their dimensionless versions. A new model for the correction factor applicable to misaligned monoliths is proposed. This model, together with existing ones, aims to provide an accurate computation of the additional pressure drop resulting from the flow exiting and entering consecutive monoliths across various gap lengths and flow rates for misalignment conditions.

2. Computational model

2.1. Description of the domain

The computational model consists of a representative section of two identical monoliths with square channels arranged in series, as shown in Fig. 2(a). The monoliths are misaligned, meaning the center axis of the

first monolith's channel does not coincide with that of the downstream monolith. Instead of that, the center axis of a channel of the first monolith lands exactly at the middle of the cross formed for the walls between channels of the second substrate. Because of that, the first monolith is directly represented by a channel, while the second monolith is represented by four fourths of a channel. Fig. 2(b) and (c) show the frontal and lateral view of such a configuration.

The domain is symmetric; hence, it is possible to reconstruct the entire substrate by expanding the selected section vertically and horizontally. Exceptions are the channels located at the external boundary of the substrate that are not perfectly square. Those represent just a small percentage of the total because a typical monolith has thousands of channels. Channel 1 had $L_1 = 16.8$ mm as length, while channel 2 had $L_2 = 6$ mm. These lengths are sufficient to prevent boundary effects in the results. The channel diameter was $D = 1.2$ mm for both monoliths. The gap between substrates, δ , varies from $0.1D$ to $20D$, and wall thickness, w , moved from 0.16 to 0.80 mm, according to the analyzed case.

The flow enters the first channel with a prescribed velocity according to the analyzed Re, passes to an open section representing the gap between substrates, and then enters the second channel that ends in a prescribed pressure outlet.

2.2. Flow model

The investigated channel Reynolds numbers were significantly lower than the critical one for square channels, so the flow was assumed to be laminar. Additionally, it was taken into account that literature reports that when turbulent flow approaches a monolith with channels with sub-critical Re, the turbulence quickly decays and the flow regime turns to laminar (Cornejo et al., 2018a, 2017). Careful should be taken because, according to published literature, turbulence can be generated when the flow leaves monolith channels (Cornejo et al., 2018b, 2020c; Hettel et al., 2019). Due to that, Large Eddy Simulation (LES) was used to corroborate the flow regime assumption. This is described in more detail in Section 2.4.

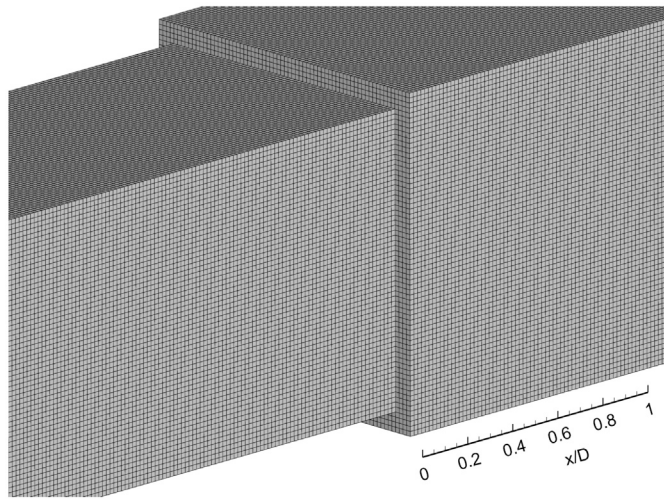
Conservation and momentum equations for the laminar steady condition were:

$$\frac{\partial(\rho u_i)}{\partial x_i} = 0 \quad (7)$$

$$\frac{\partial(\rho u_i u_j)}{\partial x_j} = -\frac{\partial p}{\partial x_i} + \frac{\partial}{\partial x_j} \left[\mu \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right) - \frac{2}{3} \frac{\partial u_i}{\partial x_i} \delta_{ij} \right] \quad (8)$$

Table 1Ranges analyzed of every variable. w is expressed in mm.

Variable	Range
Channel Reynolds	50, 100, 150, 200, 250, 300, 320
Gap length (δ/D)	0.12, 0.24, 0.48, 0.60, 0.72, 0.96, 1.20, 6.00, 12.00, 24.00
Wall thickness (w)	0.16, 0.32, 0.48, 0.64, 0.80
Void fraction (ϕ) [†]	0.36, 0.43, 0.51, 0.62, 0.78

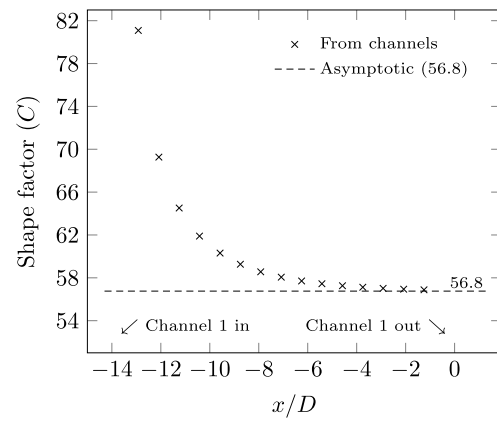
[†] Computed from the wall thickness and channel diameter.**Fig. 3.** View of the mesh. End of Channel 1 and beginning of the gap region.

As for the boundary conditions, a flat velocity profile was given to the inlet of Channel 1. The magnitude of the velocity changed according to the desired channel Re in every case (see Table 1). The inner walls and rear face of Channel 1 were set as no-slip surfaces. The left, right, top and bottom boundaries of the gap between channels were considered symmetry planes. For Channel 2, which is divided into four sections, the inner walls and frontal face were also set as no-slip boundaries, while the top, bottom left and right faces represented the center plane of the rest of the channels, hence, considered symmetry planes. Finally, the four faces at the end of Channel 2 were outlets with a prescribed pressure of 100 kPa. The operating fluid was atmospheric air at 100 kPa and 20 °C with $\rho = 1.225 \text{ kg/m}^3$ and $\mu = 1.7894 \times 10^{-5} \text{ Pa}\cdot\text{s}$.

The computational model was implemented using ANSYS Fluent 2023R2 software (2023a), which uses the finite volumes approach (FVM) (Malalasekera and Versteeg, 2007), in a x2 AMD EPYC ROME 7452 CPU HPC machine, using a total of 64 physical cores. The pressure velocity coupling was solved with the SIMPLE algorithm (Patankar, 1980), and for the momentum term, a second-order upwind scheme was used (ANSYS Fluent, 2023b). Scaled residuals below 10^{-6} and a stable pressure drop were used as stop conditions. The latter was selected because it is a relevant variable for this study and is sensitive to grid quality.

2.3. Mesh quality and validation

The domain was discretized into a series of fully orthogonal hexahedral control volumes. The aspect ratio between the edges of the control volumes was one; hence, the mesh was Cartesian. A view of the discretized domain can be seen in Fig. 3. The mesh quality was investigated assuming the most critical conditions, which were a gap size of $\delta/D = 20$, a wall thickness of $w = 0.16 \text{ mm}$ and 320 as the channel Reynolds number. For that case, the results from the two grids were compared. The first grid, or “normal”, had 1 206 600 control volumes while the second, or “refined”, had 4 826 400. The refined grid, which had four times more control volumes, was obtained by decreasing the element size homogeneously in all directions.

**Fig. 4.** Validation.

The simulation with the two grids ran until the previously mentioned stop condition. The pressure drop obtained from the normal and refined grid was 50.84 Pa and 50.76 Pa. That represents a difference of less than 0.2%, therefore, the study continued with the normal grid.

To validate the results, the friction factor's shape constant was used. It is well established in the classical literature that the pressure drop for laminar flow in ducts can be computed from the Hagen-Poiseuille expression, as in Equation (9) (White, 2009).

$$\Delta p = \frac{\Delta x}{D} \cdot f \cdot \frac{1}{2} \cdot \rho \cdot u_c^2 \quad (9)$$

The friction factor f can be written as C/Re , and Re is $(\rho \cdot u_c \cdot D/\mu)$. Replacing that into Equation (9) and rearranging, leads to the mathematical formula for C shown in Equation (10):

$$C = 2 \frac{\Delta p \cdot D^2}{\Delta x \cdot \mu \cdot u_c} \quad (10)$$

The term C is a constant that depends on the shape of the channel. When having a laminar regime, for square channels with fully developed flow, it is 56.80 (Bergman et al., 2011). Equation (10) was used to compute C along of Channel 1. The pressure drop (Δp) was calculated from consecutive segments of length D . Additionally, Δp was calculated as the difference in mass flow-weighted total pressure drop at the beginning and end of each segment. The resulting values for C are shown in Fig. 4.

It should be remarked that in the domain, $x=0$ marks the beginning of the gap between monoliths. That is, Channel 1, which is 16.8 mm long, runs from $x/D = -14$ to 0. It can be seen in Fig. 4 that C progressively decreases along the channel as the flow develops, as expected theoretically (Wilcox, 1993). The final value for C was 56.88, with an error close to 0.14% with respect to the expected 56.80. Such an accuracy is sufficient for the purpose of this work.

2.4. Confirming the flow regime

It is known that turbulence may arise when flow exits the first channel or even when it collides with the walls of the second channel downstream (Cornejo et al., 2018c). Such a phenomenon has been studied in single channels or for perfectly aligned channels but has not been investigated in a configuration like the one in this paper. For that reason, in addition to the performed in Section 2.3, a corroboration of the flow regime was also carried out. The boundary conditions were the same as in the validation section. LES was used as a flow model since it is suitable for predicting transitions in the flow regime. The simulation conditions were the same as those for the validation of the results. That is, $\text{Re} = 320$, $\delta/D = 20$ and $w = 0.16 \text{ mm}$. There was a difference in the boundary conditions of the gap section and Channel 2. Instead of using symmetry, the outer boundaries were set as periodic, mirroring the top-bottom and left-right faces in order to capture flow unsteadiness.

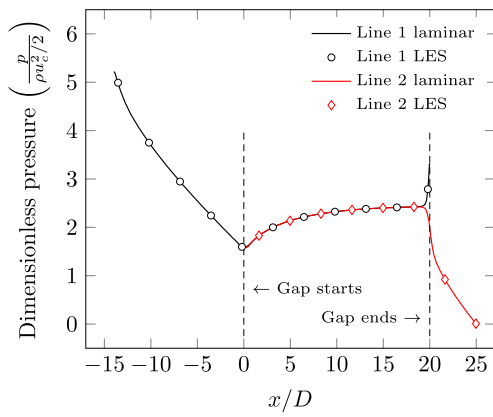


Fig. 5. Dimensionless pressure for laminar and LES results. Line 1 is the centerline of Channel 1 and line 2 is the centerline of Channel 2.

Given that LES is unsteady and the flow motion is separated into a resolved and sub-grid scale, a special notation is assumed in Equation (11) to (15). In them, the variables with an overline are those for the resolved scale after applying the size filter (Berselli et al., 2006). Mass and momentum equation balances were:

$$\frac{\partial \rho}{\partial t} + \frac{\partial(\rho \bar{u}_i)}{\partial x_j} = 0 \quad (11)$$

$$\frac{\partial \bar{u}_i}{\partial t} + \bar{u}_j \frac{\partial \bar{u}_i}{\partial x_j} = -\frac{1}{\rho} \frac{\partial \bar{p}}{\partial x_i} + \frac{\partial}{\partial x_j} \left(\nu \frac{\partial \bar{u}_i}{\partial x_j} + \frac{1}{\rho} \frac{\partial \tau_{ij}}{\partial x_j} \right) \quad (12)$$

The term τ_{ij} was computed as follows:

$$\tau_{ij} = \frac{2}{3} \cdot \rho k_{sgs} \cdot \delta_{ij} - 2C_k \cdot k^{1/2} \cdot \Delta_f \cdot \bar{S}_{ij} \quad (13)$$

The term in Equation (13) is the subgrid-scale kinetic energy, computed according to the DKE model (Kim and Menon, 1995a; Menon and Kim, 1996). In this subgrid model, k_{sgs} is a conservative quantity that responds to the following balance:

$$\frac{\partial(\rho k_{sgs})}{\partial t} + \frac{\partial(\rho \bar{u}_j k_{sgs})}{\partial x_j} = -\tau_{ij} \frac{\partial \bar{u}_i}{\partial x_j} - C_\epsilon \rho \frac{k_{sgs}^{3/2}}{\Delta_f} + \frac{\partial}{\partial x_j} \left(\frac{\mu_{sgs}}{\sigma_k} \frac{\partial k_{sgs}}{\partial x_j} \right) \quad (14)$$

where

$$\mu_{sgs} = k \cdot \rho \cdot k_{sgs}^{1/2} \cdot \Delta_f \quad (15)$$

More details about the LES and subgrid model, including the parameters and subfunctions, can be found in Kim and Menon (1995b); Kim et al. (1997); Menon et al. (1996).

To obtain more details on the resolved side of the flow, for LES, the refined grid, with 4826400 control volumes, was used. As the inlet boundary condition for turbulence, k_{sgs} was set as zero, and no turbulence generators were used. This is because the objective of this simulation is to check if turbulence is triggered by the flow leaving and entering the channels. The time step size was manipulated to keep a CFL number below unity, leading to $dt \approx 10 \mu s$. The schemes used were bounded central differencing for momentum and second-order implicit for time. Both as implemented in ANSYS Fluent 2023R2 (2023b). The problem was initialized with the inlet conditions and the stop criterion for every time step was having scaled residuals smaller than 10^{-4} . The simulation ran until it converged to a steady solution, corroborating the laminar steady flow regime assumption. The results from LES agreed with those from the steady laminar simulation in terms of pressure drop through the domain and asymptotic shape factor (C) in Channel 1.

Dimensionless static pressure along two lines of the domain obtained from LES and laminar models are compared in Fig. 5. The figure shows overlapped curves for both flow models. It is also seen, as expected, that pressure in line 1 decreases non-linearly in the first part of Channel 1,

and then the curve becomes linear as the flow develops. Entering the gap between substrates, the flow area suddenly expands. Consequently, velocity progressively decreases, and static pressure increases. Then, line 1 reaches the frontal face of Channel 2 and disappears. The flow collides that face with the respective increasing of the static pressure. Regarding line 2, it corresponds to the centerline of Channel 2, hence, it appears right at the beginning of the gap. It coincides with line 1 in the gap zone because there is no radial variation of static pressure in that part of the domain. Opposite to line 1, line 2 does not collide with walls. It enters Channel 2, where there is a marked reduction of the flow area, increasing flow velocity and suddenly decreasing pressure.

Results from both channels agree and are consistent with theoretical expectations in terms of the absolute value of the shape factor in the developed zone and the trend in static pressure. They serve as evidence confirming the choice of a laminar flow regime.

3. Results and discussion

The results analyzed involve all the combinations between 7 channel Reynolds, 10 gap lengths and 5 wall thicknesses, making a total of 350 simulations. The exact values for those variables are shown in Table 1. Channel Reynolds and wall thicknesses are typical values for monolith applications. Regarding the gap length, small and large values were considered, paying special attention to $\delta/D < 1$. For additional reference, the resulting void fractions were also reported in Table 1. Those were obtained as $\phi = D^2/(D+w)^2$.

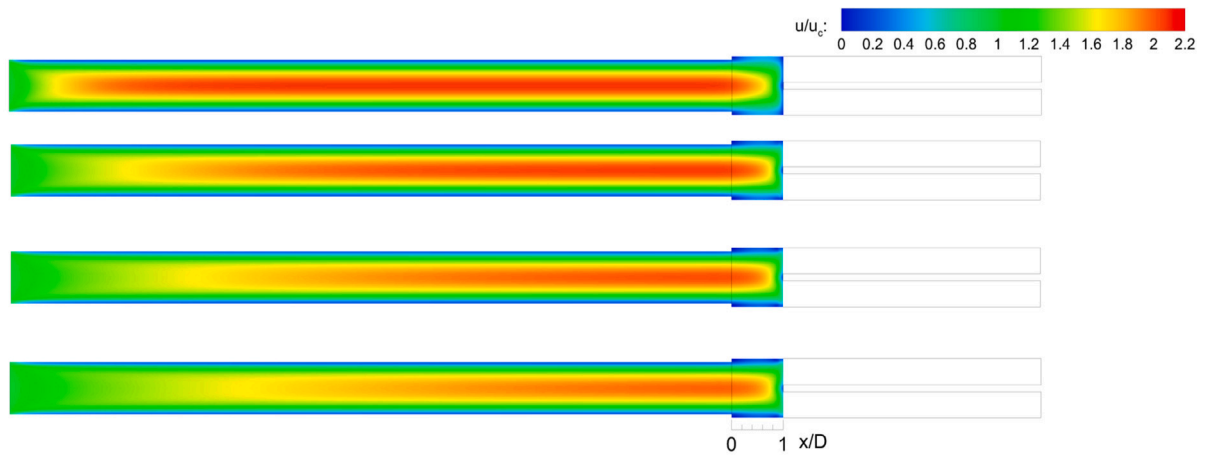
3.1. Exploratory analysis

An initial qualitative analysis of flow through misaligned channels is performed. Fig. 6 shows the effect of increasing Re at a constant gap length, while Fig. 7 illustrates the effect of increasing the gap length at a constant Re. Three planes were selected for each simulation to observe the fluid dynamics through channels. The three planes are described below:

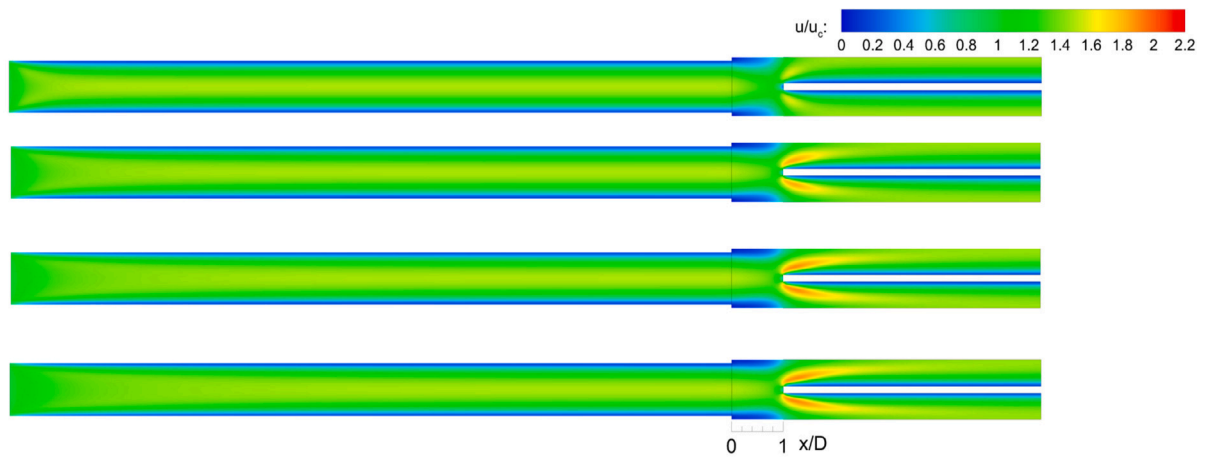
- First plane, at $z=0$. Placed at the center of Channel 1, this plane offers a view of the flow as it exits the first channel and enters the gap. It coincides with the partition wall between the quarters of Channel 2 and thus is only visible from the inlet of Channel 1 to the end of the gap zone. It allows us to observe how the flow leaving Channel 1 with the highest velocity encounters its first physical obstruction because of the misaligned channels.
- Second plane, at $z=0.28D$. Positioned halfway between the center planes of Channel 1 and Channel 2, this plane cuts through three-quarters of the way toward the left wall of the quarter-channel 2a (see Fig. 2b). It spans the entire domain, providing a view of the change in the flow trajectory and potential recirculating zones.
- Third plane, at $z=0.56D$. Aligned with the center of Channel 2, this plane captures the symmetry plane at the right boundary of the gap, along with the center of channels 2a and 2c (see Fig. 2b). It is particularly informative of the flow after fully entering the second channel, post-gap.

The results in Fig. 6 and 7 underline the significant impact of structural alignment on the flow dynamics in series-arranged monolith channels. Data from previous research regarding a perfectly aligned configuration indicates a predictable behavior as the flow moves undisturbed from one channel to the next, promoting an uninterrupted velocity profile with minimal disturbances dominated by viscosity. The misalignment investigated in this paper introduces a more complex fluid behavior due to the impact of the outflow of the first channel on the frontal face of the channel downstream, triggering flow separation and collision dynamics. The shortened gap in the misalignment configuration accentuates these phenomena, offering less space for pressure recovery

(a) Plane at $z=0D$ (coincides with the center plane of Channel 1)



(b) Plane at $z=0.28D$ (halfway between the center plane of Channel 1 and Channel 2)



(c) Plane at $z=D$ (coincides with the center plane of Channel 2)

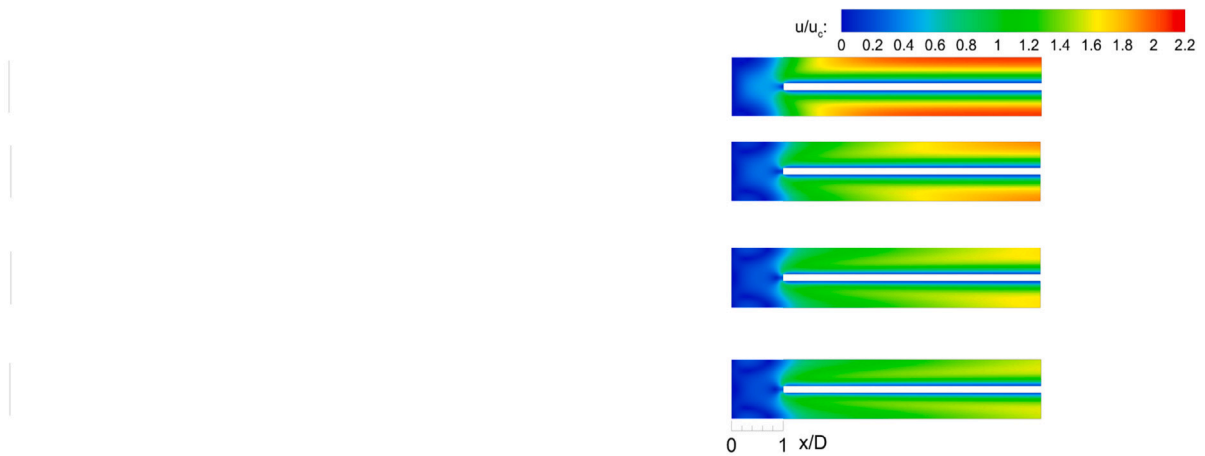


Fig. 6. Effect of increasing Re at a constant gap length of $\delta/D=0.1$. From top to bottom Re are 50, 150, 250 and 320.

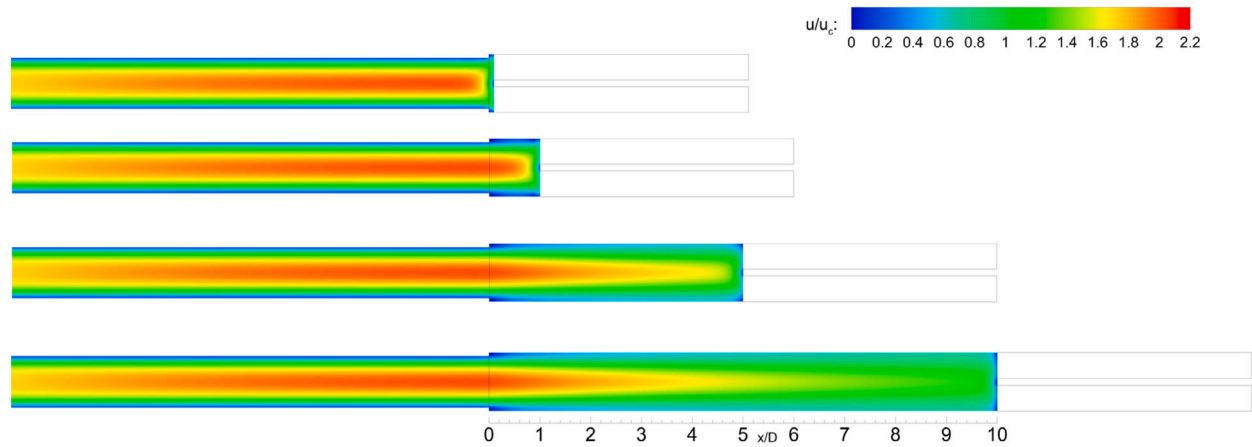
or velocity profile development before the flow undergoes a quick redirection entering the subsequent channel.

Higher Reynolds numbers in misaligned channels exacerbate these effects, making the boundary layer more prone to separation and increasing backpressure at the entrance of Channel 2. On the other hand, it seems in Fig. 6 that the zone upstream of Channel 2 affected by the viscous effects from its solid walls is independent of Re . The behavior of the boundary layer is particularly noteworthy in both Fig. 7 and 6. In misaligned channels, the boundary layer is subject to detachment and subsequent reattachment due to the direct collision with the

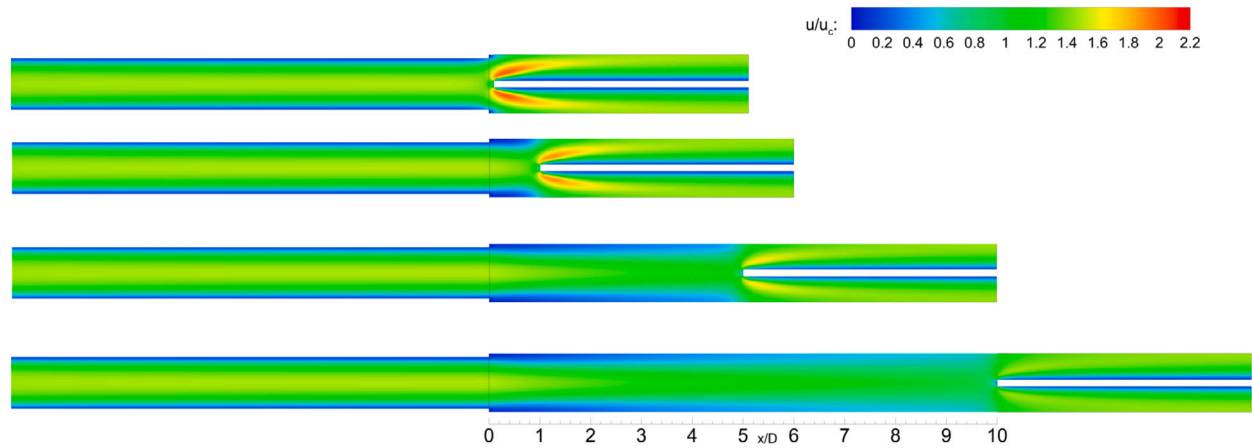
channel walls, influencing the pressure drop across the transition zone. The latter phenomenon is enhanced by a higher Re and a shorter gap length.

Fig. 8 shows a comparative analysis of the velocity and pressure fields at the entrance of the second monolith channel for short and long gaps. In the case with a short gap (Figs. 8(a) and (c)), the velocity field indicates that the flow exiting the first monolith directly collides with the wall of the second monolith channel, exhibiting energy dissipation upon impact. The collision results in localized regions of high pressure near the entrance of the second channel, as shown in the pressure field.

(a) Plane at $z=0D$ (coincides with the center plane of Channel 1)



(b) Plane at $z=0.28D$ (halfway between the center plane of Channel 1 and Channel 2)



(c) Plane at $z=D$ (coincides with the center plane of Channel 2)

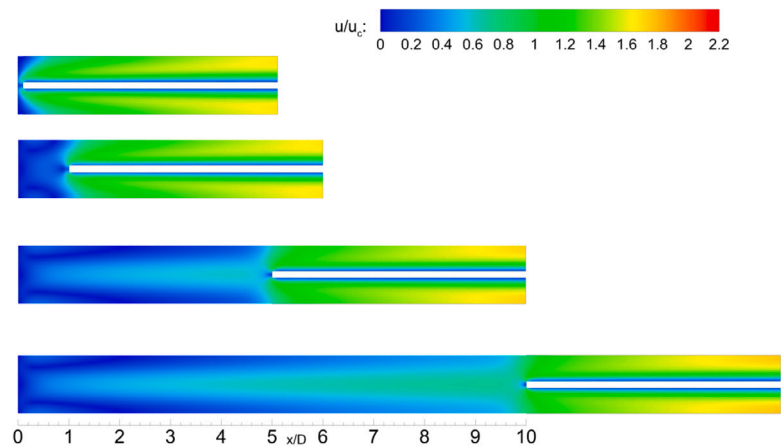


Fig. 7. Effect of increasing the gap length at a constant Re of 250. Gap lengths are $0.1D$, $1D$, $5D$ and $10D$ from top to bottom.

The velocity profile in this short gap case shows a lack of development as the flow transitions abruptly from the first to the second monolith, providing insufficient space for the flow to stabilize before entering the second monolith, potentially increasing losses.

In contrast, for the case with a large gap (Fig. 8(b) and (d)), the velocity field indicates a smoother transition of the flow from the gap into the second monolith channel. The flow has sufficient space to develop and stabilize within the gap, reducing the intensity of the collision with the wall of the second monolith. The pressure field shows a more uniform distribution with fewer high-pressure zones, indicating a more smooth flow transition and lower losses than the short gap case.

An interesting aspect to consider is how the misalignment influences the pressure drop in the downstream channel. There is an effect on the inlet velocity profile, and consequently on the hydraulic entrance length and overall pressure drop. While longer channels would be required for a thorough investigation, previous studies have explored this topic (Cornejo et al., 2019, 2020b). These studies show that a concave inlet velocity profile with peaks, similar to those obtained in this paper, affects the local friction factor and pressure drop. They also highlight that different velocity profiles do not produce substantially different pressure drops among them, but there is a noticeable difference when compared to a completely flat profile. Given that the conditions in those studies

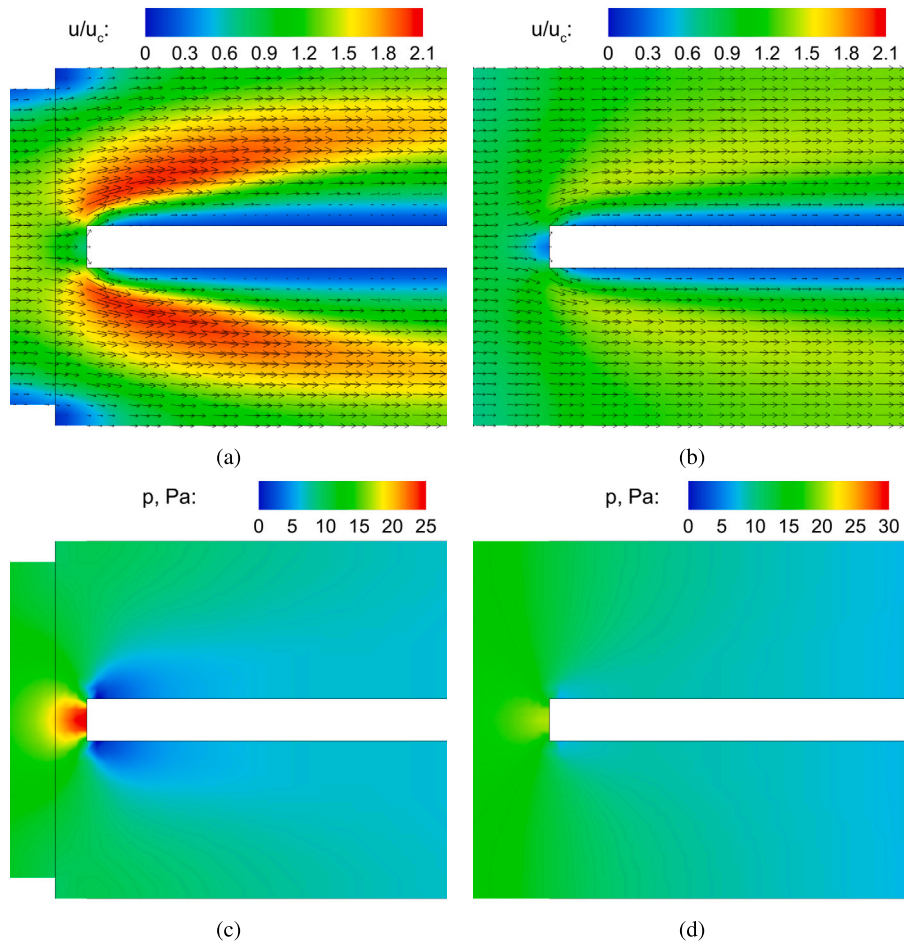


Fig. 8. Velocity and pressure fields at the entrance of the second monolith channel for two gap lengths. (a) Velocity field for $\delta/D = 0.1$. (b) Velocity field for $\delta/D = 10$. (c) Pressure field for $\delta/D = 0.1$. (d) Pressure field for $\delta/D = 10$.

are similar to those in this one, the results should apply here as well. However, further investigation considering the whole system, including the gap and downstream channels, is required to fully address this point.

3.2. Data analysis and model development

This section presents the data collected from simulations that span all the conditions shown in Table 1. The data is the pressure drop ($\Delta p_{i/o}$) from the beginning to the end of the gap between the monolith channels computed using Equation (16). In the equation, p_T is the total pressure (static plus dynamic) and A is the cross-section area at a plane at the exit of Channel 1 (P_1) and the inlet of Channel 2 (P_2). Pressure is mass flow weighted to keep consistency with the momentum balance given the change in flow profile and flow area.

$$\Delta p_{i/o} = \frac{\int_A \rho \cdot u \cdot p_T \cdot dA}{\int_A \rho \cdot u \cdot dA} \Big|_{P_1} - \frac{\int_A \rho \cdot u \cdot p_T \cdot dA}{\int_A \rho \cdot u \cdot dA} \Big|_{P_2} \quad (16)$$

Fig. 9 shows the pressure drop as a function of the scaled gap length (δ/D) for various Reynolds numbers and void fractions. It is possible to observe some trends in the figure. At very short gaps ($\delta/D < 0.4$), the pressure drop seems higher across all Reynolds numbers and void fractions, in line with the mentioned interaction effects between closely spaced monolith channels. As the gap length increases, the pressure drop decreases and stabilizes, confirming more independent flow behavior between the channels with increased spacing. The curves stabilize at longer gaps, indicating less interaction effects and pressure drops dominated by individual channel characteristics. Lower void fractions result in higher pressure drops compared to higher ones due to increased flow

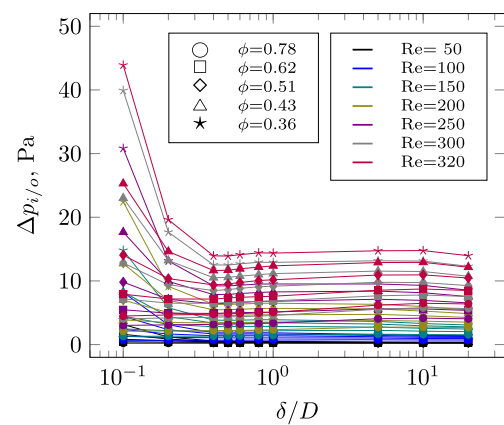


Fig. 9. Pressure drop in the gap between substrates for several void fractions (ϕ), channel Re , and scaled gap lengths (δ/D).

resistance in channels with proportionally smaller flow areas and collision zones. This effect is more pronounced at short gaps. The sharp decline in pressure drop at small gap lengths followed by a plateau suggests a transition of the phenomenon dominating losses. Higher Reynolds numbers correspond to higher losses due to increased dynamic pressure, as expected.

Fig. 10 provides an analysis of the pressure drop as a function of the scaled gap length (δ/D) for the extreme values of Re and void fractions

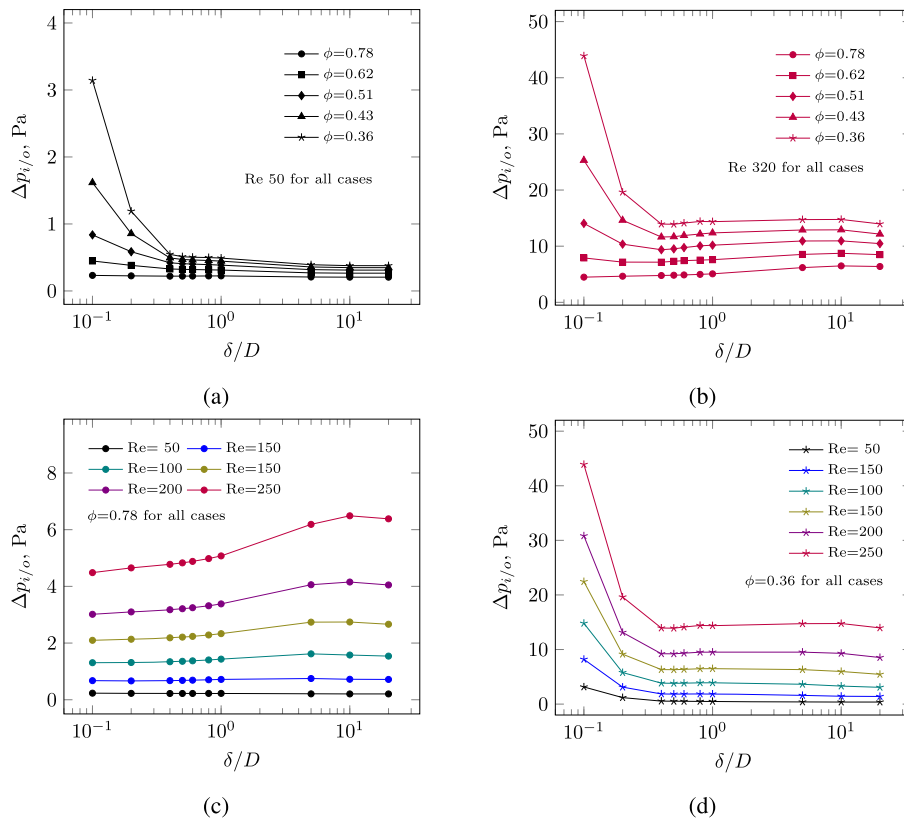


Fig. 10. Pressure drop versus scaled gap length (δ/D) highlighting the behavior at low and high Re and low and high void fraction.

considered in this study. Fig. 10(a) shows the data for $Re = 50$. Here, the pressure drop is relatively low and remains consistent for different void fractions, with a decline as the gap length increases. This indicates that the impact of void fraction on pressure drop is relatively small at low flow velocities. At a high Re, as shown in Fig. 10(b), the pressure drop is significantly higher, particularly at small gap lengths, but follows a similar trend to that for lower Re.

Regarding void fraction, data in Fig. 10(c) indicate that as the Re increases, the pressure drop also increases, particularly at short gap lengths. The pressure drop stabilizes at larger gaps, but the differences between Re remain pronounced. This trend follows the opposite pattern of that when the void fraction is low, underscoring the combined effect of low void fraction and high Re in producing pressure drops. It is observed in Fig. 10(d) that higher void fractions result in lower pressure drops, as expected, due to the reduced flow resistance within the monolith channels.

In Fig. 10(b) and 10(c), the observed trend where the pressure drop decreases with increasing gap length and then slightly increases for larger gaps might be explained by the interplay between flow resistance and secondary flow effects. Initially, as the gap length increases, the flow resistance decreases significantly because the flow obstruction is reduced, leading to a lower pressure drop. However, at larger gap lengths, secondary flow effects, such as boundary layer detachment and reattachment, begin to play a role. These effects may cause slight increases in the pressure drop because the flow dynamics become more complex, resulting in additional pressure losses. The slight increase in pressure drop for larger gaps is challenging to explain due to the involvement of multiple factors, including flow around solid collisions with the frontal face of the substrate and variations in the flow area. Each of these factors impacts the pressure drop to different extents, and the dominant phenomenon may change with void fraction and gap length, leading to complex trends. Further research is needed to provide a more precise explanation for these observations.

After analyzing the effect of the main variables on the pressure drop, a model that can fit the observed data is proposed. To do that, first, the variables are nondimensionalized.

In line with Equation (6), the curves for $\Delta p_{i/o}$ are scaled by their respective asymptotic value, which should tend to $(\Delta p_i + \Delta p_o)$, obtaining the dimensionless pressure drop $\Delta p_{i/o}^*$ that equates the correction factor. This can be done with Equation (17).

$$f_{\delta} = \Delta p_{i/o}^* = \frac{\Delta p_{i/o}}{\Delta p_i + \Delta p_o} \quad (17)$$

A new expression for δ^* for the misaligned case appeared as necessary since wall thickness plays a role in such a scenario. The choice of the expression came from fitting the values of a , b , c and d in Equation (18). The variables δ , D , and Re were included because they are part of the model for aligned channels. The wall thickness, w , was added because it impacts the frontal area where the outflow from the first channel collides.

$$\delta^* = \delta^a \cdot D^b \cdot w^c \cdot Re^d \quad (18)$$

To reduce the range of possible solutions, the values for a , b , c and d were restricted to the range from -2.5 to 2.5 in increments of 0.5. This range was based on typical values of such coefficients in boundary layer, jet flow and flow around objects problems. Additionally, a was fixed to 1 because of its physical meaning, and the sum of a , b and c was forced to be 0 in order to maintain nondimensionality. The results gave $b = 1$, $c = -2$ and $d = 0.5$. Finally, to relate the dimensionless gap with that for the aligned case, the expression was multiplied by 1 in the form of $(D \cdot Re)/(D \cdot Re)$, leading to Equation (19) after rearranging.

$$\delta^* = \delta \cdot D \cdot \frac{1}{w^2} \cdot Re^{1/2} \cdot \frac{D}{D} \cdot \frac{Re}{Re} = \left(\frac{\delta}{D \cdot Re} \right) \left[\left(\frac{D}{w} \right)^2 \cdot Re^{3/2} \right] \quad (19)$$

That mathematical treatment made it possible to propose Equation (20). The appearance of the term $(D/w)^2$ makes sense because it directly

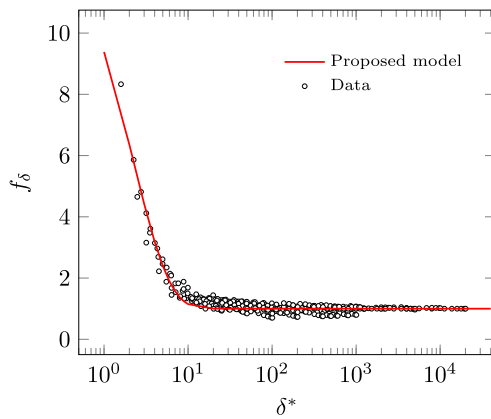


Fig. 11. Dimensionless pressure drop versus dimensionless gap length for various Reynolds numbers and void fractions.

Table 2
Model parameters and goodness of the fit.

Parameter	Value
k_c	$13.1208 \pm 0.9205^\dagger$
τ_c	$0.4478 \pm 0.0225^\dagger$
R^2	0.9314
R^2 -adjusted	0.9312
RMSE	0.1774

[†] From the 95% confidence interval

relates to the void fraction of the substrate. A misalignment modifier is explicitly included. When the channels are aligned, γ is 0, so the term simplifies to the dimensionless gap length used in previously existing models for aligned channels (Cornejo, 2021). For misaligned channels, γ is 1, so the modifier actually participates. This extends the applicability of the nondimensionalization to both aligned and misaligned cases.

$$\delta^* = \underbrace{\left(\frac{\delta}{D \cdot \text{Re}}\right)}_{\text{Aligned } x^+} \underbrace{\left[\left(\frac{D}{w}\right)^2 \cdot \text{Re}^{3/2}\right]^\gamma}_{\text{Misalignment modifier}} \quad (20)$$

Using the dimensionless variables $\Delta p_{i/o}^*$ and δ^* , a successful collapse of all the curves for the pressure drop curves against gap size is achieved. The result from applying Equation (17) and (20) to pressure drop and gap length, shown in Fig. 11, suggests that the chosen nondimensionalization is useful for generalizing the observed phenomena, at least for most of the practical applications.

Next, an equation that models the correction factor f_δ is proposed. The development of this equation considers the established model for aligned channels in Equation (6), which exhibits an exponential increase from 0 to 1 as the dimensionless gap length δ^* increases. Observing Fig. 11, an exponential decrease from the initial to an asymptotic value of 1 is noticed. This inverse exponential behavior contrasts with the aligned case but suggests that a similar mathematical structure could be applicable.

$$f_\delta = 1 + k_c \cdot \exp(-\tau_c \cdot \delta^*) \quad (21)$$

Same as in the case for aligned channels, the two parameters k_c and τ_c are obtained through nonlinear regression. It should be noted that Equation (21) can be linearized to $\ln(\Delta p_{i/o} - 1) = \ln(k_c) - \tau_c \cdot \delta^*$, hence, the regression has a unique solution (Montgomery, 1996; Box et al., 2005). The result of the curve fitting is summarized in Table 2.

Figs. 12 and 13 compare the proposed model against the data. Fig. 12 shows the convergence of the pressure drop curves, demonstrating the effectiveness of the nondimensionalization and proposed model to describe the data in the whole range analyzed. Additionally, Fig. 13 shows the correlation between the measured and predicted values, further

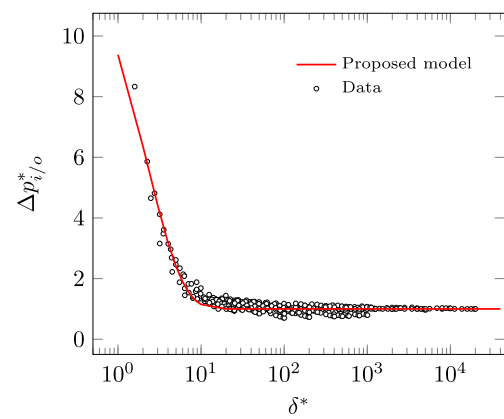


Fig. 12. Predicted dimensionless pressure drop using the proposed model versus dimensionless gap length compared to data.

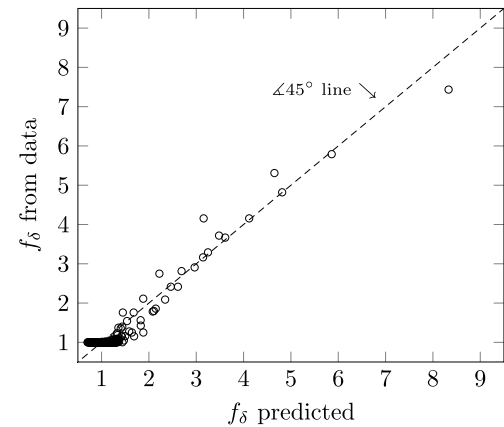


Fig. 13. Correlation plot of predicted using the proposed model versus computed from data correction factor f_δ .

Table 3
Parameter values for the aligned and misaligned cases.

Case	γ	k_c	τ_c	Negligible effects if
Aligned [†]	0	1	45.92	$\delta^* \geq 10^{-1}$
Misaligned	1	13.12	0.4478	$\delta^* \geq 10^1$

[†] From Cornejo (2021).

complementing the goodness of fit. It can be seen from this figure that the majority of data points closely align along the 45° line. Some lack of sensitivity is observed near unity and for the point at (8,7). Additionally, the points describe a slightly curvy shape, but these errors are tolerable for practical applications, provided that this is the first model of its kind trying to describe the phenomenon.

Finally, due to their mathematical similarity, a general expression for both aligned and misaligned channels can be proposed by combining Equation (6) and Equation (21). The resulting model is shown in Equation (22).

$$f_\delta = 1 + (-1)^{1-\gamma} \cdot k_c \cdot \exp(-\tau_c \cdot \delta^*) \quad (22)$$

Replacing δ^* according to Equation (20), a more explicit form is obtained:

$$f_\delta = 1 + (-1)^{1-\gamma} \cdot k_c \cdot \exp\left\{-\tau_c \left(\frac{\delta}{D \cdot \text{Re}}\right) \left[\left(\frac{D}{w}\right)^2 \cdot \text{Re}^{3/2}\right]^\gamma\right\} \quad (23)$$

The model is useful for both aligned and misaligned cases when using the correct set of parameters. It gives the value of the correction factor to be used in Equation (5). Table 3 summarizes parameter values.

To prevent the most negative scenario of increased extra pressure drop, it is important to maintain sufficient separation between substrates. The results show that the extra pressure drop does not depend on the absolute distance δ , but on the dimensionless one δ^* , which is affected by Re , w , D and alignment condition, as shown in Equation (20). A recommended value for δ^* from which negative effects due to misalignment can be neglected is shown in Table 3.

4. Conclusions

This study successfully introduces a new model for predicting additional pressure drops in misaligned monolith catalyst substrates arranged in series. The findings highlight a notable impact of misalignment on pressure drops, particularly in closely spaced monoliths. However, it is important to note that the model is currently limited to monoliths misaligned by half of a channel diameter. Additionally, no rotation of the monoliths is considered. Addressing such limitations would require a significantly longer set of simulations, which is beyond the scope of the present exploratory study.

It is concluded that for the studied misalignment, the extra pressure drop and correction factor differ significantly from that for the aligned scenario. These factors are significantly influenced by the void fraction, Reynolds number, and gap length. Higher Reynolds numbers and lower void fractions result in increased pressure drops due to greater dynamic pressure and frontal collision. Shorter gap lengths lead to higher pressure drops due to increased interaction effects between fluid and solid.

Dimensionless variables are introduced to generalize the phenomenon in a typical range of conditions for monolith applications. Based on them, a unified correction factor model, applicable to both aligned and misaligned channels, extends existing frameworks. By incorporating a misalignment modifier, the model predicts the correction factor across different gap lengths, flow rates and void fractions, enhancing its practical applicability. Minor discrepancies, such as the slight curvature in the correlation plot, are acknowledged but acceptable for practical applications.

Experimental validation of the proposed model is highly recommended, along with an exploration of the effect on the pressure drop for different channel shapes.

This study also acknowledges that misalignment affects the pressure drop in the downstream channels. It is expected that non-homogeneous inflow profiles due to misalignment cause variations in local friction factor and pressure drop. However, further investigation with longer channels is needed to fully understand this effect.

Nomenclature

C	Shape factor
D	Channel hydraulic diameter, m
f	Darcy friction factor, -
f_1, f_2	Model parameters, -
f_δ	Correction factor, -
γ	Misalignment term
k_c	Model parameter, -
k_{sgs}	Sub-grid turbulence kinetic energy, m^2/s^2
w	Substrate wall thickness, m
p	Pressure, Pa
p_T	Total pressure, Pa
Δp_c	Pressure drop inside of a substrate, Pa
Δp_i	Pressure drop when entering a substrate, Pa
Δp_o	Pressure drop when leaving a substrate, Pa
$\Delta p_{i/o}$	Combined effect of Δp_o and Δp_i for substrates in series, Pa
Re	Reynolds number, -
t	Time, s
u	Velocity magnitude, m/s
u_c	Channel velocity, m/s
δ	Length of the gap between substrates, m

δ_{ij}	Kronecker delta, -
ϕ	Substrate void fraction, -
ρ	Fluid density, kg/m^3
τ_c	Model parameter, -
μ	Fluid viscosity, Pa·s
μ_{sgs}	Sub-grid turbulence viscosity, Pa·s

CRediT authorship contribution statement

Ivan Cornejo: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgements

Financial support provided by ANID/R23F0001 is acknowledged. The author is also grateful to Gonzalo Garretón for his discussions on the preliminary results and to Consuelo Reinao for her significant contribution to the graphical abstract.

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